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Remote Services: SMB/Windows Admin Shares

Other sub-techniques of Remote Services (8)

Adversaries may use **Valid Accounts** to interact with a remote network share using Server Message Block (SMB). The adversary may then perform actions as the logged-on user.

SMB is a file, printer, and serial port sharing protocol for Windows machines on the same network or domain. Adversaries may use SMB to interact with file shares, allowing them to move laterally throughout a network. Linux and macOS implementations of SMB typically use Samba.

Windows systems have hidden network shares that are accessible only to administrators and provide the ability for remote file copy and other administrative functions. Example network shares include **C\$**, **ADMIN\$**, and **IPC\$**. Adversaries may use this technique in conjunction with administrator-level **Valid Accounts** to remotely access a networked system over SMB.^[1] to interact with systems using remote procedure calls (RPCs),^[2] transfer files, and run transferred binaries through remote Execution. Example execution techniques that rely on authenticated sessions over SMB/RPC are **Scheduled Task/Job, Service Execution**, and **Windows Management Instrumentation**. Adversaries can also use NTLM hashes to access administrator shares on systems with **Pass the Hash** and certain configuration and patch levels.^[3]

Procedure Examples

ID	Name	Description
C0025	2016 Ukraine Electric Power Attack	During the 2016 Ukraine Electric Power Attack, Sandworm Team utilized net use to connect to network shares. ^[4]
S0504	Anchor	Anchor can support windows execution via SMB shares. ^[5]
G0007	APT28	APT28 has mapped network drives using Net and administrator credentials. ^[6]
C0051	APT28 Nearest Neighbor Campaign	During APT28 Nearest Neighbor Campaign, APT28 leveraged SMB to transfer files and move laterally. ^[7]
G0022	APT3	APT3 will copy files over to Windows Admin Shares (like ADMIN\$) as part of lateral movement. ^[8]
G0050	APT32	APT32 used Net to use Windows' hidden network shares to copy their tools to remote machines for execution. ^[9]
G0087	APT39	APT39 has used SMB for lateral movement. ^[10]
G0096	APT41	APT41 has transferred implant files using Windows Admin Shares and the Server Message Block (SMB) protocol, then executes files through Windows Management Instrumentation (WMI). ^{[11][12]}
G0143	Aquatic Panda	Aquatic Panda used remote shares to enable lateral movement in victim environments. ^[13]
G1043	BlackByte	BlackByte used SMB file shares to distribute payloads throughout victim networks, including BlackByte ransomware variants during wormable operations. ^{[14][15][16]}
S1180	BlackByte Ransomware	BlackByte Ransomware uses mapped shared folders to transfer ransomware payloads via SMB. ^[17]
S0089	BlackEnergy	BlackEnergy has run a plug-in on a victim to spread through the local network by using PsExec and accessing admin shares. ^[18]
G0108	Blue Mockingbird	Blue Mockingbird has used Windows Explorer to manually copy malicious files to remote hosts over SMB. ^[19]
S1063	Brute Ratel C4	Brute Ratel C4 has the ability to use SMB to pivot in compromised networks. ^{[20][21][22]}
G0114	Chimera	Chimera has used Windows admin shares to move laterally. ^{[23][24]}
G1021	Cinnamon Tempest	Cinnamon Tempest has used SMBexec for lateral movement. ^[25]
S0154	Cobalt Strike	Cobalt Strike can use Window admin shares (C\$ and ADMIN\$) for lateral movement. ^{[26][27]}
S0608	Conficker	Conficker variants spread through NetBIOS share propagation. ^[28]
S0575	Conti	Conti can spread via SMB and encrypts files on different hosts, potentially compromising an entire network. ^{[29][30]}
C0029	Cutting Edge	During Cutting Edge, threat actors moved laterally using compromised credentials to connect to internal Windows systems with SMB. ^[31]
G0009	Deep Panda	Deep Panda uses net.exe to connect to network shares using net use commands with compromised credentials. ^[32]
S0659	Diavol	Diavol can spread throughout a network via SMB prior to encryption. ^[33]
S0038	Duqu	Adversaries can instruct Duqu to spread laterally by copying itself to shares it has enumerated and for which it has obtained legitimate credentials (via keylogging or other means). The remote host is then infected by using the compromised credentials to schedule a task on remote machines that executes the malware. ^[34]
S0367	Emotet	Emotet has leveraged the Admin\$, C\$, and IPC\$ shares for lateral movement. ^{[35][36]}
G1016	FIN13	FIN13 has leveraged SMB to move laterally within a compromised network via application servers and SQL servers. ^[37]
G0061	FIN8	FIN8 has attempted to map to C\$ on enumerated hosts to test the scope of their current credentials/context. FIN8 has also used smbexec from the Impacket suite for lateral movement. ^{[38][39]}
G0117	Fox Kitten	Fox Kitten has used valid accounts to access SMB shares. ^[40]
S0698	HermeticWizard	HermeticWizard can use a list of hardcoded credentials to to authenticate via NTLMSSP to the SMB shares on remote systems. ^[41]
C0038	HomeLand Justice	During HomeLand Justice, threat actors used SMB for lateral movement. ^{[42][43]}
G0004	Ke3chang	Ke3chang actors have been known to copy files to the network shares of other computers to move laterally. ^{[44][45]}
S0236	Kwampirs	Kwampirs copies itself over network shares to move laterally on a victim network. ^[46]
G0032	Lazarus Group	Lazarus Group malware SierraAlfa accesses the ADMIN\$ share via SMB to conduct lateral movement. ^{[47][48]}

ID: T1021.002

Sub-technique of: T1021

 **Tactic:** Lateral Movement

 **Platforms:** Windows

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C0049	Leviathan Australian Intrusions	Leviathan used remote shares to move laterally through victim networks during Leviathan Australian Intrusions. ^[49]
S1199	LockBit 2.0	LockBit 2.0 has the ability to move laterally via SMB. ^{[50][51]}
S1202	LockBit 3.0	LockBit 3.0 can use SMB for lateral movement. ^[52]
S0532	Lucifer	Lucifer can infect victims by brute forcing SMB. ^[53]
G1054	MirrorFace	MirrorFace has used SMB to copy malware between systems in compromised environments. ^{[54][55]}
G1009	Moses Staff	Moses Staff has used batch scripts that can enable SMB on a compromised host. ^[56]
S0039	Net	Lateral movement can be done with Net through <code>net use</code> commands to connect to the on remote systems. ^[57]
S0056	Net Crawler	Net Crawler uses Windows admin shares to establish authenticated sessions to remote systems over SMB as part of lateral movement. ^[58]
S0368	NotPetya	NotPetya can use PsExec, which interacts with the <code>ADMIN\$</code> network share to execute commands on remote systems. ^{[59][60][61]}
S0365	Olympic Destroyer	Olympic Destroyer uses PsExec to interact with the <code>ADMIN\$</code> network share to execute commands on remote systems. ^{[62][61]}
C0048	Operation MidnightEclipse	During Operation MidnightEclipse, threat actors used SMB to pivot internally in victim networks. ^[63]
C0014	Operation Wocao	During Operation Wocao, threat actors used Impacket's smbexec.py as well as accessing the C\$ and IPC\$ shares to move laterally. ^[64]
G0071	Orangeworm	Orangeworm has copied its backdoor across open network shares, including ADMIN\$, C\$WINDOWS, D\$WINDOWS, and E\$WINDOWS. ^[46]
G1040	Play	Play has used Cobalt Strike to move laterally via SMB. ^[65]
S0029	PsExec	PsExec, a tool that has been used by adversaries, writes programs to the <code>ADMIN\$</code> network share to execute commands on remote systems. ^[61]
S1242	Qilin	Qilin can embed a copy of PsExec within its payload and place it in the %Temp% directory under a randomly generated filename. ^{[66][67]}
S1212	RansomHub	RansomHub can use credentials provided in its configuration to move laterally from the infected machine over SMBv2. ^[68]
S1187	reGeorg	reGeorg has the ability to tunnel SMB sessions. ^[69]
S0019	Regin	The Regin malware platform can use Windows admin shares to move laterally. ^[70]
S1073	Royal	Royal can use SMB to connect to move laterally. ^[71]
S0446	Ryuk	Ryuk has used the C\$ network share for lateral movement. ^[72]
G0034	Sandworm Team	Sandworm Team has copied payloads to the <code>ADMIN\$</code> share of remote systems and run <code>net use</code> to connect to network shares. ^{[4][73]}
S0140	Shamoon	Shamoon accesses network share(s), enables share access to the target device, copies an executable payload to the target system, and uses a Scheduled Task/Job to execute the malware. ^[74]
C0024	SolarWinds Compromise	During the SolarWinds Compromise, APT29 used administrative accounts to connect over SMB to targeted users. ^[75]
G1046	Storm-1811	Storm-1811 has attempted to move laterally in victim environments via SMB using Impacket. ^[76]
S0603	Stuxnet	Stuxnet propagates to available network shares. ^[77]
G0028	Threat Group-1314	Threat Group-1314 actors mapped network drives using <code>net use</code> . ^[78]
G1022	ToddyCat	ToddyCat has used locally mounted network shares for lateral movement through targated environments. ^[79]
G0010	Turla	Turla used <code>net use</code> commands to connect to lateral systems within a network. ^[80]
G1047	Velvet Ant	Velvet Ant has transferred tools within victim environments using SMB. ^[81]
G0102	Wizard Spider	Wizard Spider has used SMB to drop Cobalt Strike Beacon on a domain controller for lateral movement. ^{[82][83]}
S0672	Zox	Zox has the ability to use SMB for communication. ^[84]
S0350	zwShell	zwShell has been copied over network shares to move laterally. ^[85]

Mitigations

ID	Mitigation	Description
M1037	Filter Network Traffic	Consider using the host firewall to restrict file sharing communications such as SMB. ^[86]
M1035	Limit Access to Resource Over Network	Consider disabling Windows administrative shares.
M1027	Password Policies	Do not reuse local administrator account passwords across systems. Ensure password complexity and uniqueness such that the passwords cannot be cracked or guessed.
M1026	Privileged Account Management	Deny remote use of local admin credentials to log into systems. Do not allow domain user accounts to be in the local Administrators group multiple systems.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0530	Multi-Event Detection for SMB Admin Share Lateral Movement	AN1468	An SMB-based remote file share access followed by lateral movement actions such as remote service creation, task scheduling, or suspicious process execution on the target host using ADMIN\$ or C\$ shares.

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